

DV Pipeline with Social Media Analytics Platform- GCP - Use case

Tasks done in brief:

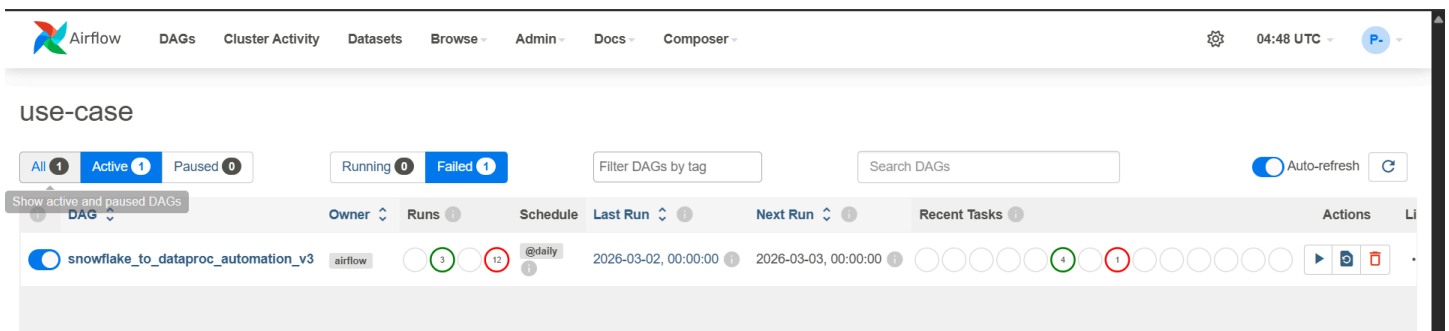
Component	Role in your Use Case
Snowflake	The primary data transformation engine.
GCS	The "landing zone" for both the export and the import.
Dataproc	The ETL worker that formatted data for BigQuery.
BigQuery	The high-performance backend for your reporting.
Airflow	The conductor making sure the timing was perfect.
Looker Studio	The "Face" of the project for the end-users.

Tasks Done

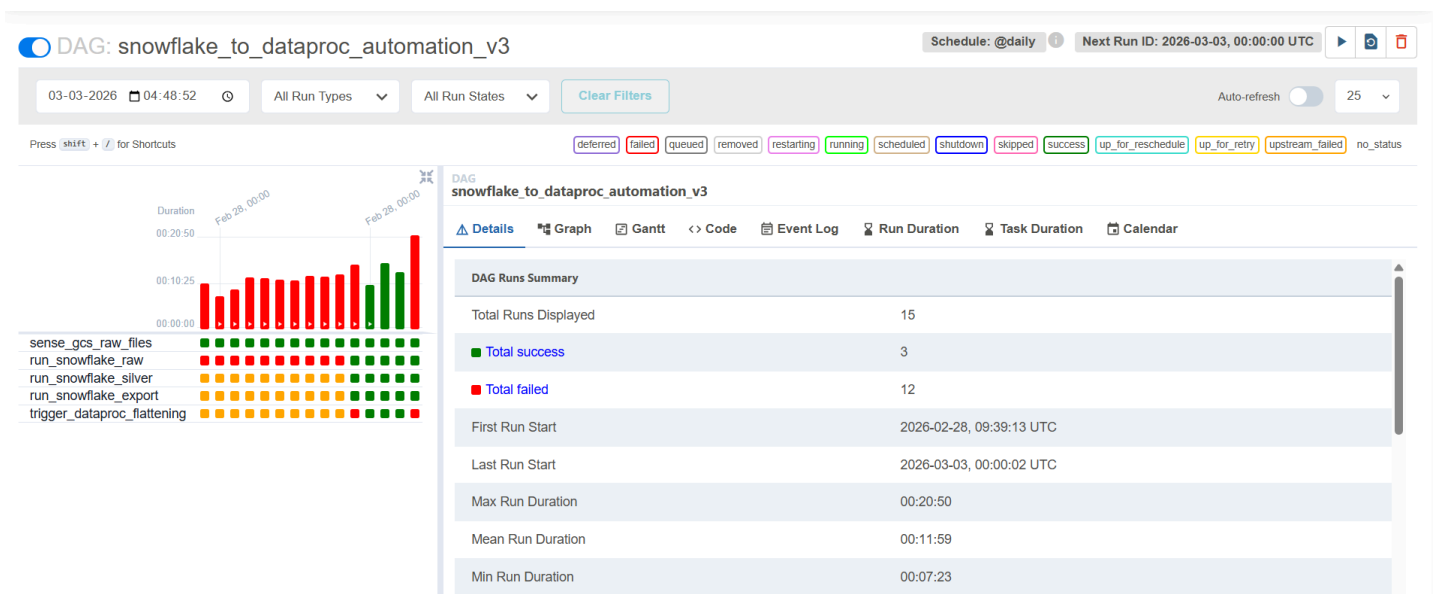
1. End-to-End Orchestration (Airflow/Cloud Composer)

- **Automated Workflow Design:** Developed a Python-based Airflow DAG to automate the data lifecycle from landing to final BigQuery ingestion.
- **Event-Driven Triggering:** Implemented a `GCSObjectExistenceSensor` to ensure the pipeline only executes when new raw employment data arrives in the GCS bucket, optimizing resource usage.
- **Error Handling & Idempotency:** Configured the DAG with retry logic and ensured tasks are idempotent (using `OVERWRITE` logic) to prevent data duplication in the event of a restart.

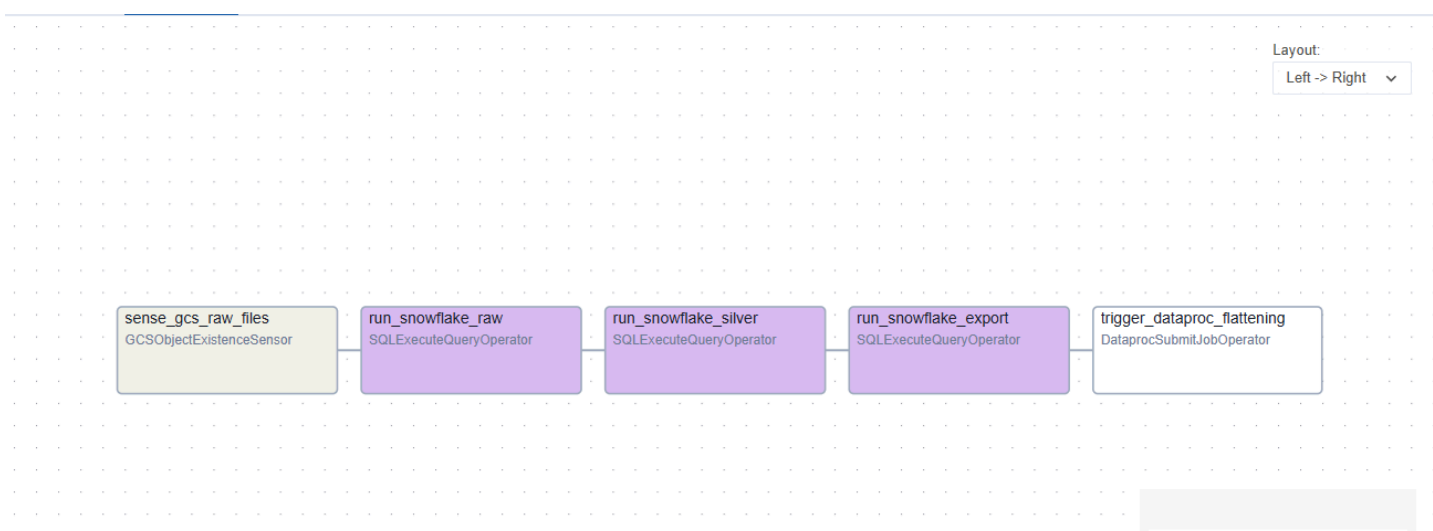
DAGS



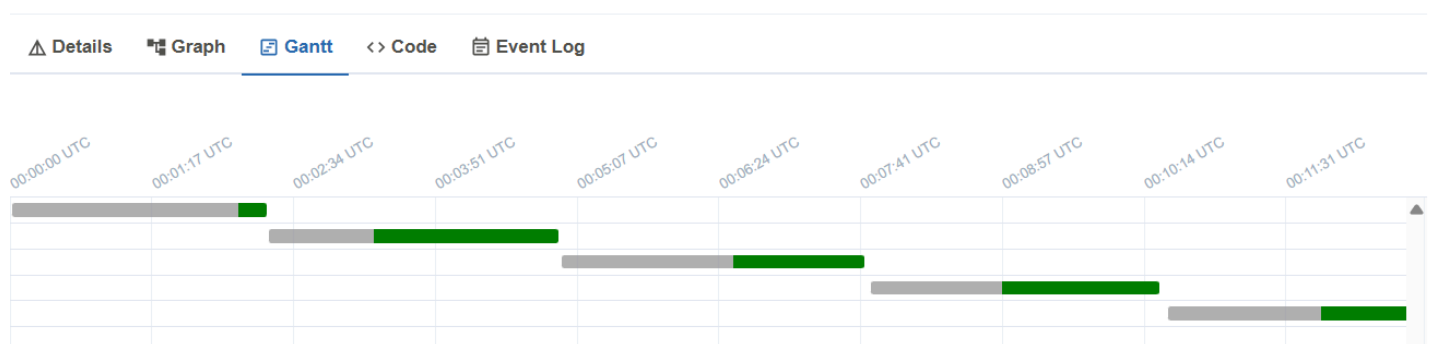
DAG Job UI



Job flow



Gantt chart



Connection specially the Snowflake one

List Connection

Search ▾

 Actions ▾ 

Record Count: 6

☐	Conn Id ↕	Conn Type ↕	Description ↕	Host ↕	Port ↕	Is Encrypted ↕	Is Extra Encrypted ↕
☐	  airflow_db	postgres		airflow-sqlproxy-service.composer-system.svc.cluster.local	3306	True	False
☐	  bigquery_default	google_cloud_platform				False	True
☐	  google_cloud_datastore_default	google_cloud_platform				False	True
☐	  google_cloud_default	google_cloud_platform				False	True
☐	  google_cloud_storage_default	google_cloud_platform				False	True
☐	  <u>snowflake_conn</u>	snowflake				True	True

2. Cloud Run Function

- The `split-social-media-excel` Cloud Function processes the `employee.csv` file by splitting the dataset into multiple files based on defined logic and uploads the generated files to the curated folder in the Google Cloud Storage bucket.
- The Bucket `social-media-dropshipping-analytics-land` Stores the `/raw` and `/curated` files

3. Snowflake Integration & Stored Procedures

- **Automated Processing:** Migrated manual SQL scripts into **Snowflake Stored Procedures** (`SP_USE_CASE_RAW_LANDING`, `SILVER`, `EXPORT`) to encapsulate business logic within the data warehouse.



```
SQL use-case_raw_landing.sql
SQL use-case-export.sql
SQL use-case-silver.sql
```

- **Secure Connection Management:** Configured the Airflow-to-Snowflake connection (`snowflake_conn`), including advanced "Extra" metadata parameters to resolve account identifier and warehouse authentication issues.
- **GCS Data Export:** Orchestrated the `COPY INTO` command to export processed records back to the GCS `curated/` folder, utilizing `OVERWRITE = TRUE` to maintain a clean data state for downstream processing.

4. Dataproc & BigQuery Ingestion

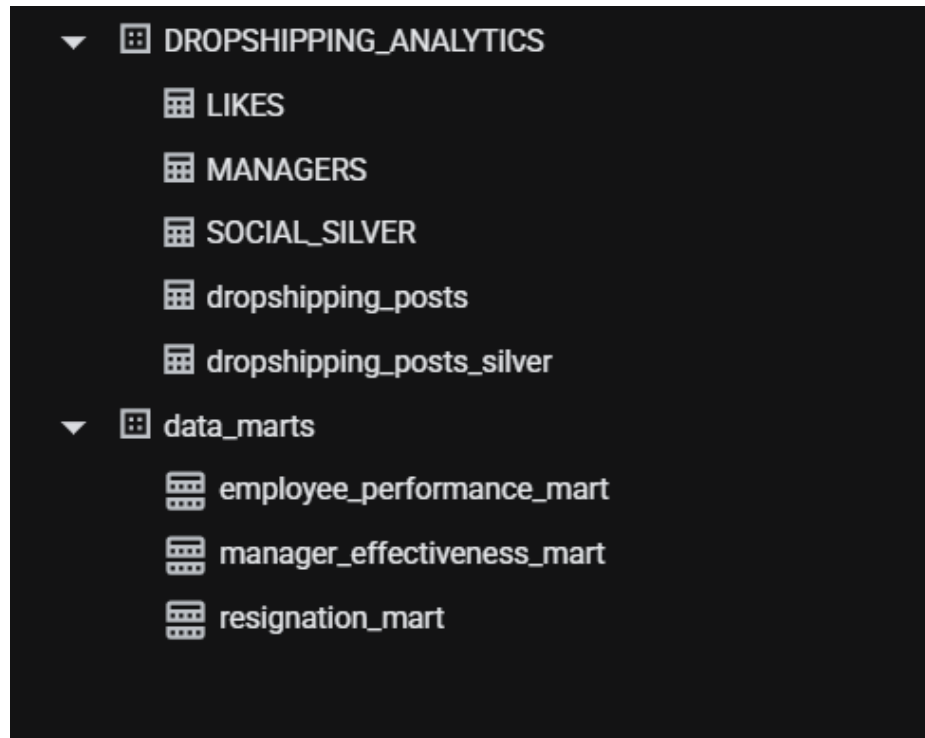
- **PySpark Job Orchestration:** Integrated the `DataprocSubmitJobOperator` to trigger a `flatten_metadata.py` script immediately following Snowflake's export.
- **Advanced Data Flattening:** Automated the transformation of complex Snowflake exports into a flattened schema suitable for high-performance analytics in BigQuery.
- **Schema Enforcement:** Ensured the final data load into BigQuery maintains integrity for future reporting.

☐	dropshipping-cluster	 Stopped	us-east1	us-east1-d	2.2.76-debian12
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☐	ebbb8aae-ab0b-4b65-8900-70d9d5afe108	 Succeeded	us-east1	PySpark	dropshipping-cluster
☐	c695ef5e-f901-45ed-a620-d6b8d8dd3bff	 Succeeded	us-east1	PySpark	dropshipping-cluster
☐	b9176874-8e49-4d45-9263-77ccf9361529	 Succeeded	us-east1	PySpark	dropshipping-cluster
☐	job-83af7f09	 Succeeded	us-east1	PySpark	dropshipping-cluster

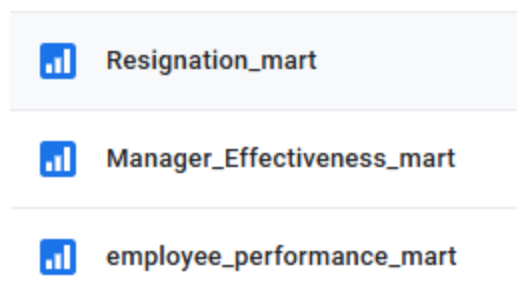
6. Analytics Data Mart Construction (BigQuery)

- **Modular Data Modeling:** Designed and implemented three specialized **Data Marts** in BigQuery to serve as the "gold" layer for reporting, ensuring data is clean, pre-aggregated, and easy to query.
- **Employee Performance Mart:** Aggregated individual KPIs, attendance records, and onsite opportunity metrics to provide a 360-degree view of employee productivity.
- **Manager Effectiveness Mart:** Synthesized team-level data, retention rates, and internal feedback scores to measure leadership impact and departmental health.
- **Resignation Mart:** Created a dedicated trend analysis table to track turnover rates, exit drivers, and average tenure, allowing HR to identify risk patterns before they escalate.



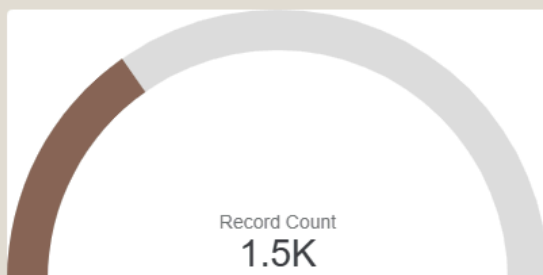
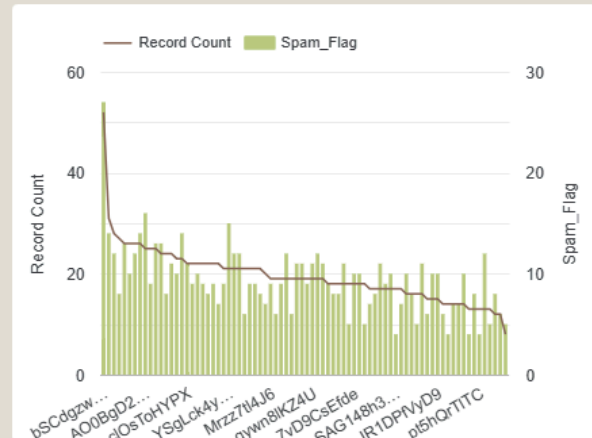
7. BI Visualization & Insight Delivery (Looker Studio)

- **Cost-Optimized Reporting:** Leveraged **Looker Studio** as the primary visualization layer, providing a powerful, no-cost alternative to expensive BI services like Looker (LookML) without sacrificing dashboard quality.



	Post_ID	Record Count
1.	bSCdgzwOhC	52
2.	UuLU5ecSWz	31
3.	0QHJYJ3afbG	28
4.	VOFaLJ9tGf	27
5.	CJylz7NhpE	26
6.	vryP1zfUnt	26
7.	p5jqj1pQfu	26
8.	1Ysljn8S8S	26
9.	iBpHtGQDA	26

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	Region_Code	Churn_...
1.	NORG	174.43
2.	EMEA	164.16
3.	LATAM	152.39
4.	APAC	144.87
5.	MEA	134.77

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Summary of the "End-to-End" Flow

1. **GCS/Snowflake:** Raw data is landed, cleaned, and transformed into Silver/Export layers.
2. **Airflow/Dataprocc:** Automation triggers the flattening of complex Snowflake JSON/CSV exports.
3. **BigQuery:** Data is partitioned into the 3 Marts described above.
4. **Looker:** Business users interact with the visual data to make hiring and retention decisions.

IAM Roles used

IAM					
Allow	Deny	Recommendations history			
☐		118804472838-compute@developer.gserviceaccount.com	Default compute service account	Editor	
				Storage Object Admin	Advanced security insight
☐		dv-pipeline-488406@appspot.gserviceaccount.com	App Engine default service account	BigQuery Data Editor	Advanced security insight
				BigQuery Job User	Advanced security insight
				Dataproc Editor	Advanced security insight
				Editor	
				Logging Admin	Advanced security insight
				Pub/Sub Publisher	Advanced security insight
				Storage Object Admin	Advanced security insight
☐		looker-connection-981@dv-pipeline-488406.iam.gserviceaccount.com	looker-connection	BigQuery Data Viewer	Advanced security insight
				BigQuery Job User	Advanced security insight
☐		looker-connection@dv-pipeline-488406.iam.gserviceaccount.com	looker-connection	BigQuery Data Viewer	Advanced security insight
☐		pipeline-orchestrator-proj-500@dv-pipeline-488406.iam.gserviceaccount.com	pipeline-orchestrator@project-id.iam.gserviceaccount.com	Eventarc Event Receiver	Advanced security insight
				Pub/Sub Lite Admin	Advanced security insight
				Storage Object Admin	Advanced security insight
☐		prateekhandekar01@gmail.com	Prateek	Advisory Notifications Viewer	Advanced security insight
				Organization Administrator	Advanced security insight
				Owner	
				Project Mover	Advanced security insight